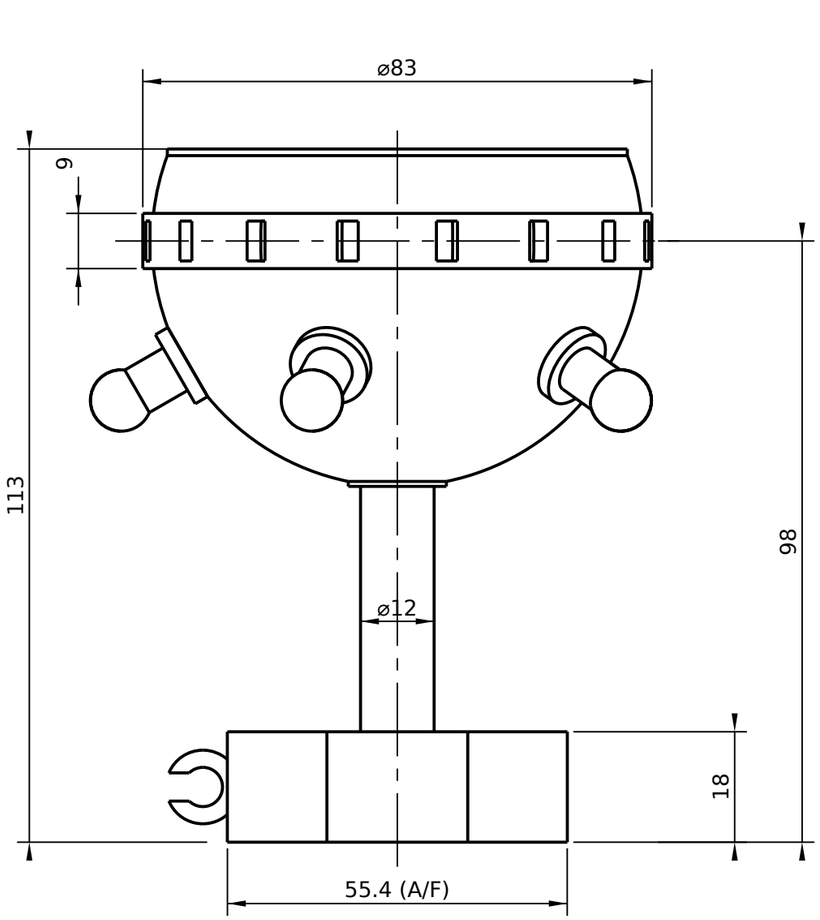

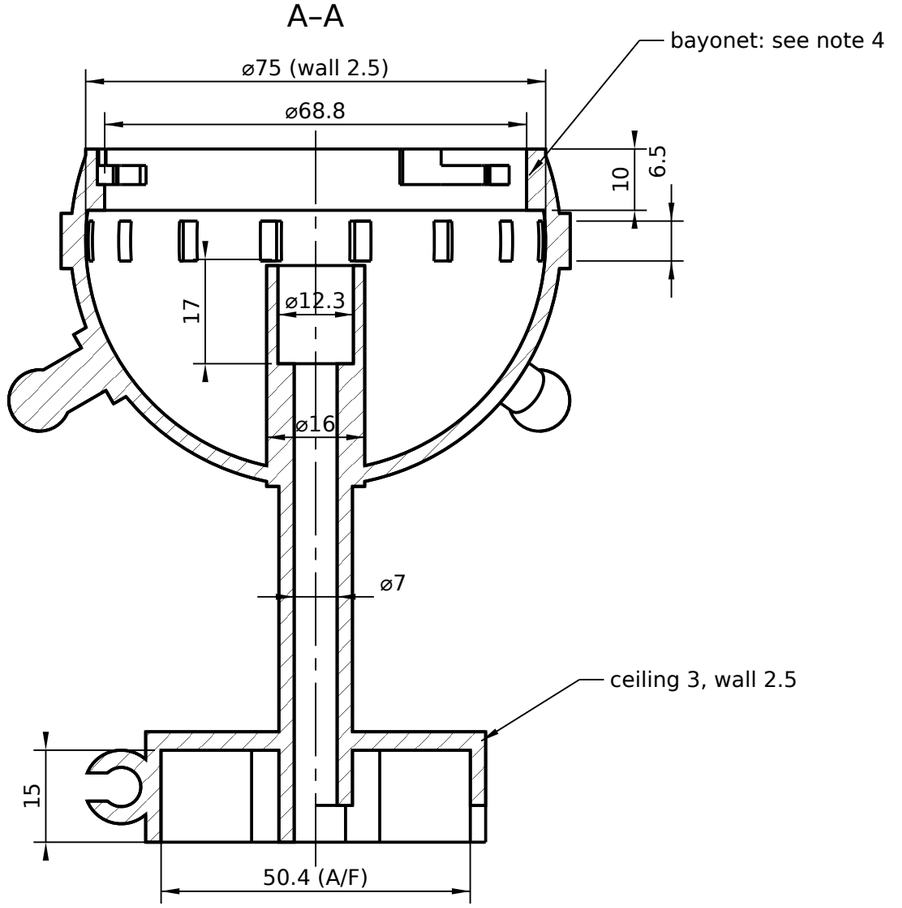

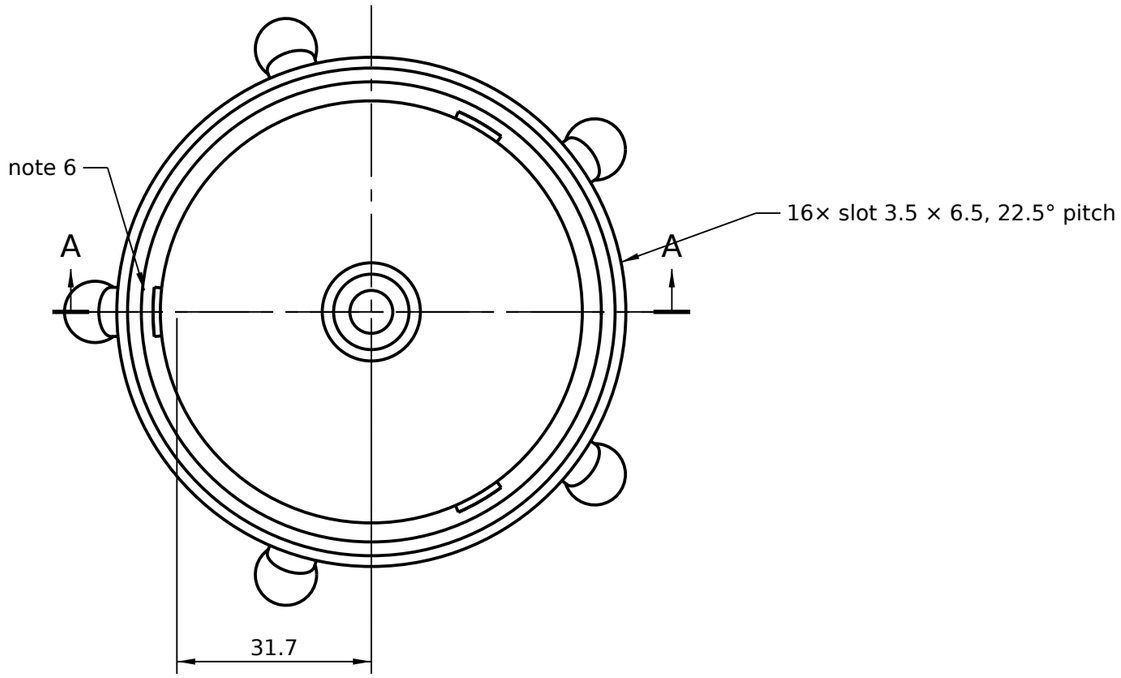

Notes

1 Print in HP MJF PA12, no supports; powder must drain from the open base.

2 Base cavity: fill with dry sand, seal flush with aquarium-safe silicone.

3 Unspecified radii/edges as modelled; reference model mine_bottom.step.

4 Bayonet: 3 L-slots at 120°, depth to ø71.2, entry 13°, turn 30° clockwise, channel height 3.1, 0.3 mm detent step at end of travel.

5 LED pocket ø12.3 fits a 12 mm WS2811 bullet pixel (push fit).

6 Probe clip ø12 × 16, bore ø6.4 for a ø6 DS18B20 probe, opening 4.6.

7 5× horn ø8 with ø10 tip, 30° below the equator, 72° pitch.

Material	Surface	Mass (approx.)	Scale
PA12 (HP MJF)	as printed, dyed black	≈ 68 g + sand	1:1
General tolerances	**Unit**	**Created by / Date**	**Approved by**
ISO 2768-m	mm	develmusa / 2026-10-26	–
Document type	**Document status**	**Owner**	**Projection**
Part drawing	Prototype	Thermomine	

Thermomine – aquarium mine thermometer

Bottom part (base, tether, lower shell)

Drawing number	Revision	Sheet
TM-001-01	**A**	1/1